Beyond Direct Generation: A Decomposed Approach to Well-Crafted Screenwriting with LLMs

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Abstract

The screenplay serves as the foundation for television production, defining narrative structure, character development, and dialogue. While Large Language Models (LLMs) offer great potential in this creative process, direct end-to-end generation approaches often fail to produce well-crafted screenplays. We argue this failure stems from forcing a single model to simultaneously master two disparate capabilities: creative narrative construction and rigid format adherence. To enable LLMs to generate high-quality screenplays, we introduce Dual-Stage Refinement (DSR), a decomposed framework that explicitly decouples creative narrative generation from format conversion. In the first stage, the framework transforms a brief outline into rich, novel-style prose. The second stage then refines this prose into a professionally formatted screenplay. This separation enables the model to specialize in one distinct capability at each stage. A significant challenge in implementing DSR is the scarcity of paired outline-to-novel data. We address this through a hybrid data synthesis strategy that combines reverse synthesis (deconstructing existing screenplays into structured inputs) and forward synthesis (generating high-quality novel-style texts as training targets). Extensive experiments show that in blind evaluations by professional screenwriters, screenplays generated by DSR achieve a 75% win rate against strong baselines like Gemini-2.5-Pro and reach 82.7% of human-level performance, demonstrating that decomposed generation architecture is highly effective for specializing LLMs in complex creative domains.

1 Introduction

The advent of LLMs brings new potential to screenplay generation[0]. LLMs can serve as creative assistants by handling time-consuming tasks such as generating plot variations and character biographies, allowing writers to focus on higher-level creative decisions. However, current general-purpose LLMs must be adapted to understand screenplay-specific narrative structures, maintain story coherence, and manage complex character development. The challenge is developing specialized AI tools that enable writers to produce higher-quality screenplays efficiently.

A screenplay differs fundamentally from a novel as a structured framework for visual storytelling. Rather than relying on prose descriptions or internal monologue, screenplays employ "showing, not telling" through precise audiovisual language—scene headings, concise action lines, and authentic dialogue. Consequently, an effective screenplay generation model must maintain narrative coherence, ensure character consistency, and adhere to strict formatting conventions.

Current research has made notable progress in controllable story generation, developing models that follow plot outlines (Rashkin et al., 2020) or incorporate commonsense knowledge (Wang et al., 2022). In screenwriting, systems like Dramatron (Mirowski et al., 2022) employ hierarchical prompt chaining.

However, these approaches focus primarily on narrative content and logical coherence. We address a more complex challenge: generating production-ready screenplays that require both creative narrative generation ("what to write") and rigid format conversion ("how to write"). Handling both skills jointly increases training complexity and often produces outputs that superficially mimic screenplay style but lack professional quality. This suggests decoupling these skills, allowing the model to focus on one task at a time.

We introduce Dual Stage Refinement (DSR), a decomposed framework that separates screenplay generation into two stages. The first stage translates high-level outlines into rich, novel-style

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[0]The terms 'screenplay' and 'script' in this work refer to scripts for episodic television, not feature films.

prose, focusing purely on storytelling. The second stage refines this prose into professionally formatted screenplays. This separation enables the model to master one distinct skill at each step.

Implementing DSR presents a data challenge: the lack of paired outline-to-novel data for training the first stage. We address this through a hybrid data synthesis strategy combining reverse synthesis (deconstructing screenplays into structured inputs) and forward synthesis (generating high-quality novel-style targets).

Blind evaluations by professional screenwriters demonstrate DSR's effectiveness: a 75% win rate against SOTA models like Gemini-2.5-Pro and Claude-Sonnet-4, reaching 82.7% of human-level performance.

Our main contributions are: **(1)** The DSR framework that decouples screenplay generation into creative narrative development and format conversion, addressing limitations of end-to-end approaches; **(2)** A hybrid data synthesis strategy combining reverse and forward synthesis to resolve data scarcity; **(3)** Comprehensive validation showing DSR significantly outperforms strong baselines in blind evaluations by professional screenwriters.

2 Related Works

Screenplay Generation. Early screenplay generation relied on retrieval-based methods (Zhu et al., 2022). Modern LLM-based frameworks like Dramatron (Mirowski et al., 2022) and IBSEN (Han et al., 2024) enable interactive co-writing but require substantial human supervision. Tian et al. (2024) found that while LLMs excel at surface-level qualities, they fail at "narrative intelligence" in end-to-end generation due to the difficulty of simultaneously managing creative construction and structural constraints. Unlike collaborative frameworks, our work tackles autonomous, high-quality generation.

Planning-based Narrative Generation. Plan-and-write approaches use sparse plans (Brei et al., 2023; Wang and Kreminski, 2024; Wang et al., 2023) or detailed outlines (Yang et al., 2023b), but these create semantic gaps between planning and execution. Sophisticated pipelines like CML-BENCH (Zheng et al., 2025), HoLLMwood (Kor et al., 2023), and R^2 (Lin et al., 2024) still rely on lossy, non-narrative representations. Iterative refinement methods like Re3 (Yang et al., 2022) attempt costly post-generation fixes.

We address these limitations by using novels as dense, narratively-complete intermediate states, decoupling creative generation (outline-to-novel) from format conversion (novel-to-screenplay).

Data Synthesis for Narrative Tasks. High-quality training data is scarce for screenplay generation. Prior work employs reverse synthesis (You et al., 2023; Huang et al., 2023; Ahuja et al., 2024) to extract plans from texts, or forward synthesis (Yang et al., 2023a; Zhu et al., 2024) using LLMs to generate training data. However, these focus on single-stage generation.

Our two-stage approach requires outline-to-novel pairs unavailable in existing datasets. We employ hybrid synthesis: reverse synthesis extracts structured inputs from screenplays, then forward synthesis generates novel-style narratives, producing the training pairs needed for our framework.

3 Methodology

This section presents DSR, our proposed framework for generating well-crafted screenplays. We begin by formally defining the screenwriting task and examining the limitations of direct end-to-end generation (Section 3.1). We then introduce the DSR framework's two-stage architecture that decouples creative narrative generation from format conversion (Section 3.2).

3.1 Task Formulation

The primary objective is to generate a screenplay scene S, based on a set of structured inputs X. The input X is a tuple $\{O, P, C, M\}$, where:

- Scene Outline (O): The scene-by-scene outline, guiding the plot's progression.
- Previous Context (P): The premise or prior context, ensuring narrative continuity.
- Character Profiles (C): Character profiles, defining their backgrounds, personalities, and relationships.
- Metadata (M): Specific instructions that guide the creative tone or content, such as "focus on the escalating conflict between Character A and B" or "this scene must end on a cliffhanger".

A fundamental challenge in this domain is the scarcity of paired training data. The available resources for our task consist solely of finalized screenplays from aired television series. We lack

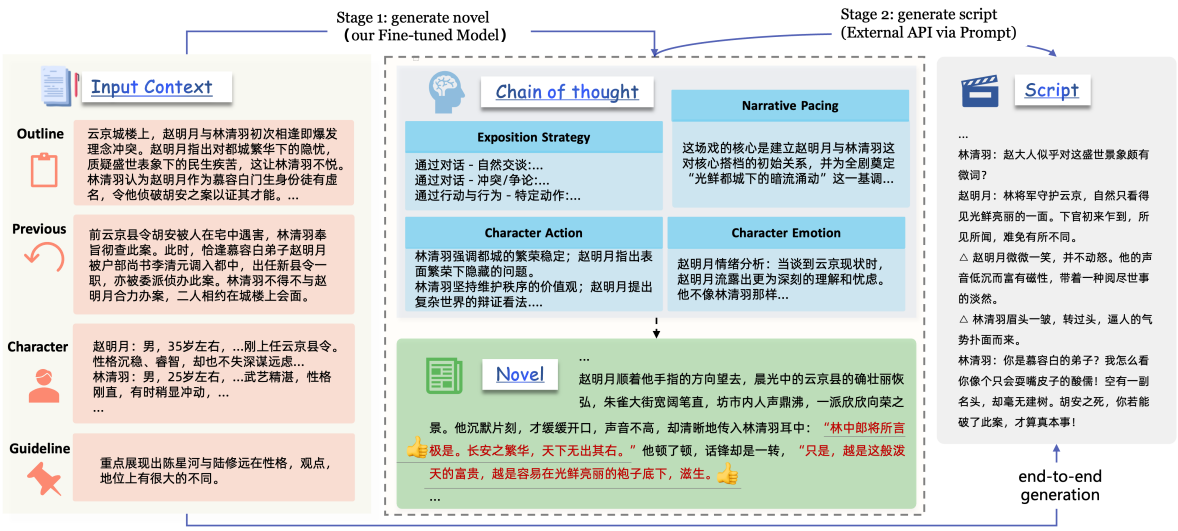

Figure 1: The proposed two-stage generation pipeline. Stage 1 focuses on creative narrative generation (Outline-to-Novel), while Stage 2 handles structural formatting (Novel-to-Screenplay).

access to the corresponding outlines or authorial notes that writers originally used to create these screenplays. This data-availability constraint necessitates the construction of our own training pairs. Our primary task, therefore, is to create a high-quality dataset $\mathcal{D} = \{(X_i, S_i)\}_{i=1}^{|\mathcal{D}|}$, where the inputs X_i are reverse-engineered to serve as plausible creative briefs for the existing screenplays S_i.

It is crucial to acknowledge that screenwriting is an open-ended creative task. For any given input X_i, there is no single "ground-truth" screenplay. Therefore, in our constructed dataset, the target screenplay S_i should be viewed not as a definitive answer, but as a high-quality reference representing one point sampled from the vast space of possible valid solutions. This reference demonstrates the desired narrative structure, character voice, and screenplay format. The model's objective is not to replicate this specific sample, but to learn the general mapping from a creative brief to a professionally valid screenplay instance.

With this formulation and the constructed dataset, a standard Supervised Fine-Tuning (SFT) approach aims to train a model M_θ with parameters θ to directly learn the conditional probability distribution $P(S|X)$. The objective is to find the parameters θ^* that maximize the total log-likelihood of the target screenplays across the dataset $\mathcal{D}$. This is formally expressed as:

$$\theta^* = \arg\max_{\theta} \sum_{(X,S)\in\mathcal{D}} \log P(S|X;\theta) \qquad (1)$$

A straightforward approach to this task would be to train a model to directly generate the final screenplay from the input context in a single step. This naive baseline is visually represented by the blue arrow labeled "end-to-end generation" in Figure 1, which depicts a direct mapping from the Input Context (comprising Outline, Previous scenes, Character profiles, and Guidelines) to the final Script, bypassing the intermediate Novel representation and the explicit Chain of Thought (CoT) reasoning process. Our preliminary experiments with this end-to-end approach revealed suboptimal results. The generated screenplays often suffer from a lack of thematic focus, out-of-character dialogue, and insufficient character development. We attribute these shortcomings to the **Task Coupling Dilemma** faced by single-stage models. Specifically, the model is forced to simultaneously master two disparate skills: **(1) Narrative Generation:** The creative task of elaborating a story from a high-level outline into rich, detailed prose (as shown in the Novel box in Stage 1). This requires imaginative expansion to design the scene's pacing, its sequential development, and the intricate chain of cause and effect that drives the story forward. The model must reason through multiple dimensions simultaneously, including Exposition Strategy, Narrative Pacing, Character Action, and Character Emotion (illustrated in the Chain of Thought boxes), to construct a coherent dramatic skeleton. **(2) Format Conversion:** The task of transforming descriptive narratives into the visual and auditory language of screenwriting (Stage 2 in Figure 1).